Le Trung Kien Nguyen

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Education

University of South Florida Expected May 2027
B.S. Mechanical Engineering | GPA: 3.84/4.00 *Tampa, FL*
- **Coursework:** Mechanics of Solids, Dynamics, Modeling of Engineering Systems, Kinematics of Machinery

Experience

VinMotion (Humanoid Robotics) Jun. 2025 – Aug. 2025
Mechanical Engineering Intern *Hanoi, Vietnam*
- Inspected **19 humanoid robot assemblies** (16 A1.6, 3 A2.1); filed failure reports, corrected SolidWorks CAD errors, and recommended design changes to senior engineers before a public exhibition.
- Built **2 full-scale 3D-printed duo-planetary actuator prototypes** with the actuator team through CAD review, tolerance and fit checks, and assembly, enabling in-house actuator testing.
- Authored A1.6/A2.1 humanoid assembly instructions, standardizing build procedures and shortening onboarding.

C&D Semiconductor Services Summer 2025
CAD Design Intern *Remote (San Jose, CA)*
- Designed **5 new custom parts** in SolidWorks for testing new **photoresist coating and developer modules**.
- Revised **100 custom parts**, equipment modules, and a **process cabinet** in SolidWorks, keeping CAD current.
- Created **engineering drawings and assemblies** for legacy equipment versions to build future design documentation.

USF SOAR – NASA Student Launch Sep. 2024 – Present
Aerostructures Engineer, Airbrake Mechanism Design *Tampa, FL*
- Improved airbrake drag by **about 200%** over **3 SolidWorks prototype iterations**, optimizing lever-arm travel, packaging, and flap geometry to reach **42 N of drag per flap** within a **4 lb system mass budget**.
- Designed a stepper-driven ACME lead-screw and lever-arm flap actuator; measured mass properties and flap angles to support **PID controller calibration** and flight-readiness testing.
- Cut damaged-fin replacement time by **about 50%** with a bolted fin-can frame; fabricated and assembled custom rocket components with hand and shop tools, resolving fit issues during integration.

Projects

Cycloidal Gearbox Actuator for Robot Joints | *SolidWorks, FDM 3D Printing, DFM* 2026 – Present
- Designed **20:1 and 30:1 cycloidal reducers** in SolidWorks for NEMA 23, targeting **10 N-m torque**, backdrivable.
- Applied DFM for FDM printing, compensating for **+/-0.4 mm printer tolerance** to balance fit, backlash, and friction heating; selected bearings and fasteners for prototype assembly.
- Developing a compact NEMA 17 variant with **two cycloidal disks on an eccentric cam**; planning an **Al 6061** version.

12-DOF Quadruped Robot, Webbed Feet | *Arduino, Raspberry Pi, C/C++* Sep. 2022 – May 2023
- Designed bio-inspired webbed feet that increased contact area **25x** (0.0002 to 0.005 m^2), reducing foot submersion in 50%-water mud from **35/21/20/10 mm to 10/6/5/3 mm**; **ISEF 2023 Finalist**.
- Sized **12 servo motors** (3 DOF/leg) via first-principles torque calculations, integrated **FSR force sensors**, and programmed **8 gaits** with **inverse kinematics**, climbing a **20-degree incline with a 1 kg payload**.

Technical Skills

CAD & Design: SolidWorks (CSWA), surface modeling, engineering drawings, GD&T, tolerance analysis, DFM/DFA
Analysis: FEA (Ansys, SolidWorks Simulation), topology optimization, stress analysis, torque and motor sizing
Robotics: Actuators, gearboxes, inverse kinematics, servo/stepper motors, sensors, Arduino, Raspberry Pi, URDF, PID
Prototyping: FDM 3D printing (Bambu Lab), assembly and fit-checking, soldering, shop tools, McMaster-Carr sourcing
Programming: Python, C/C++, MATLAB/Simulink
Certifications: Certified SOLIDWORKS Associate (CSWA); SOLIDWORKS Additive Manufacturing Associate

Awards & Leadership

Awards: USAID Science for Development Award (3rd), ISEF 2023; 1st Prize, HCMC Science Fair 2023; 1st, Technion Robotraffic 2023
Leadership: Head of Technical Faculty, Le Hong Phong Science Club; led robotics competition teams